JOSEPH KELLY

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EDUCATION

Northeastern University **Boston, MA**

Khoury College of Computer Science **GPA: 3.73/4.00**

B.S. in Computer Science, Systems Concentration, Minor in Mathematics *May 2028*

Relevant Coursework: Algorithms and Data, Fundamentals of Software Engineering, Object-Oriented Design, Introduction to Databases, Mathematical Foundations of Artificial Intelligence, Programming in C++, Foundations of Cybersecurity, Probability and Statistics, Theory of Computation, Discrete Structures, Fundamentals of Computer Science 1 & 2, Multivariable Calculus

EXPERIENCE

SLATE

Technical Operations & Automation Co-op *Jul 2026 – Present*

- Engineered and deployed a Python/FastAPI QA system to automate daily validation of standardized healthcare claims against original source files across 23 administrators; tested 1,678 cases spanning 8,219 files (6.5 GB) across diverse formats.
- Built specialized parsers for spreadsheets, delimited files, and fixed-width pharmacy records with 459 configurable validation rules; accelerated CSV parsing 10x using PyArrow and reduced test suite runtime 32%, preserving results across 469 regression tests.
- Developed a 3-stage LLM pipeline to identify financial exposure in healthcare plans and stop-loss contracts across 168 audit questions, incorporating OCR, Pydantic-validated outputs, secondary LLM verification, and 74 passing tests with pytest.
- Created an internal Python tool to de-identify protected health information, enabling privacy-conscious AI workflows.
- Led a team-wide overhaul of client project planning, introducing user stories and Conditions of Satisfaction to clarify requirements, strengthen project tracking, and align deliverables with client expectations throughout client projects.

Alactive **London, UK**

Mobile Frontend Engineer *Nov 2024 – Dec 2024*

- Delivered 3 SwiftUI iOS features through 4 peer-reviewed pull requests with an early-stage startup engineering team.
- Refactored 15 deprecated SwiftUI components to reduce technical debt and improve maintainability across the iOS app.

PROJECTS

MailCart – Founder & Full-Stack Developer *Mar 2026 – Present*

React, FastAPI, PostgreSQL, Stripe, Clerk, OpenAI API

- Built and deployed a full-stack SaaS for personalized email outreach using React, FastAPI, and PostgreSQL; integrated Gmail and Microsoft Graph APIs and recorded 477 successful provider sends across 21 campaigns through connected mailboxes.
- Engineered asynchronous per-recipient email sending with spreadsheet-driven personalization, conditional HTML, inline images, attachments, and live NDJSON progress streaming; implemented cancellation, retry handling, and persisted send outcomes.
- Developed 47 REST API endpoints and 13 PostgreSQL tables, integrating Clerk authentication, Stripe subscriptions and team-seat billing, AI-assisted email generation, server-enforced usage limits, Stripe billing webhooks, and admin analytics.
- Maintained 495 passing automated tests (387 backend, 108 frontend) across 30 test files, covering API integrations, authentication, billing, email rendering, and 20 shared cross-language personalization scenarios aligning UI previews with sent messages.

NU Insights – AI Course & Degree Planner *Jan 2026 – Present*

React, Flask, SQLite, OR-Tools, OpenRouter API, Docker, Railway

- Engineered an OR-Tools CP-SAT semester planner enforcing prerequisites, corequisites, credit limits, and degree requirements; preserved completed and pinned courses, accommodated co-op terms, and verified planning logic with 52 automated tests.
- Built and deployed a React/Flask application backed by indexed SQLite; scraped 6,034 courses across 162 subjects and requirements for 395 undergraduate program variants, adding four archived catalog editions for year-specific degree planning.
- Integrated 4 LLM providers (OpenAI, Anthropic, Gemini, and OpenRouter) for catalog-grounded explanations and personalized elective ranking. Validated course selections and added deterministic fallback for malformed responses, timeouts, and rate limits.

LEADERSHIP & COMMUNITY

Northeastern Men's Club Volleyball **Boston, MA**

Treasurer *Apr 2026 – Present*

- Manage team finances and prepare monthly financial reports tracking expenses, tournament costs, player dues, and budgets.
- Coordinate travel logistics, flights, hotels, tournament fees, and fundraising to control team expenses and minimize player dues.

Experience Camps **Oakland, ME**

Cabin Counselor *Aug 2024 – Present*

- Mentor grieving children at overnight camp, providing emotional support and ensuring safety during medical emergencies.

TECHNICAL KNOWLEDGE

Languages: Python, JavaScript, Java, SQL, C++, TypeScript, Swift, Kotlin, HTML, CSS

Frameworks/Libraries: React, FastAPI, Flask, OR-Tools, SwiftUI, pandas, PyArrow, Pydantic, BeautifulSoup, pytest

Databases/Infrastructure: PostgreSQL, SQLite, Docker, Vercel, Railway, Git, GitHub, Linux/Unix

APIs/Services: OpenAI, Anthropic, Gemini, OpenRouter, Gmail, Microsoft Graph, Stripe, Clerk